

REAL ESTATE ANALYTICS

Ames Housing Market: Data-Driven Pricing Insights

Empowering investors with high-precision valuation models to eliminate market uncertainty.

The Valuation Challenge

Why traditional appraisals often miss the mark.

Valuation Pain Points

- ❗ **Market Uncertainty:** Overpaying for assets or underselling high-potential properties.
- ❗ **Complexity Overload:** Sifting through 80+ house features is impossible for human agents.
- ❗ **Outlier Distortion:** Unique properties "skew" average prices, leading to bad data.



Our **Scientific** Approach



Noise Removal

We use Z-score filtering to remove extreme outliers, ensuring our model learns from clean, reliable market data.



Log Transformation

Home prices aren't linear. We transform data to capture percentage-based value growth for more natural predictions.



Lasso Logic

The Lasso model acts as a "Smart Filter," automatically muting irrelevant data to focus on the 10% of features that drive 90% of price.

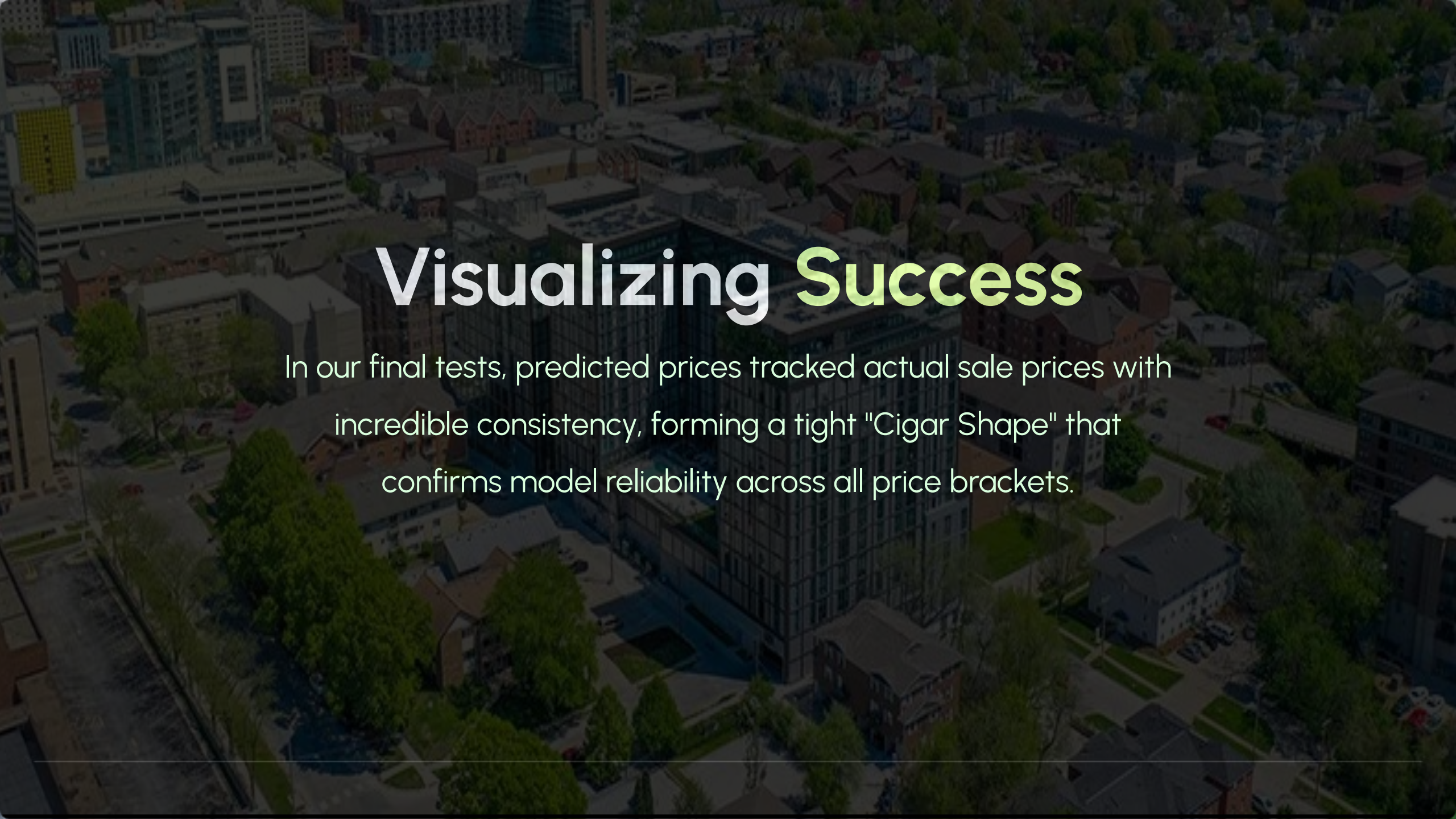
Unmatched Precision

89.3%

Variance Captured (R^2)

The Gold Standard

Our final model successfully explains nearly 90% of all price fluctuations in the Ames market. This level of accuracy provides a massive competitive advantage for rapid decision-making.

An aerial photograph of a city, likely New York City, showing a dense urban landscape with various buildings, streets, and green spaces. The image is overlaid with a semi-transparent dark layer to make the text stand out.

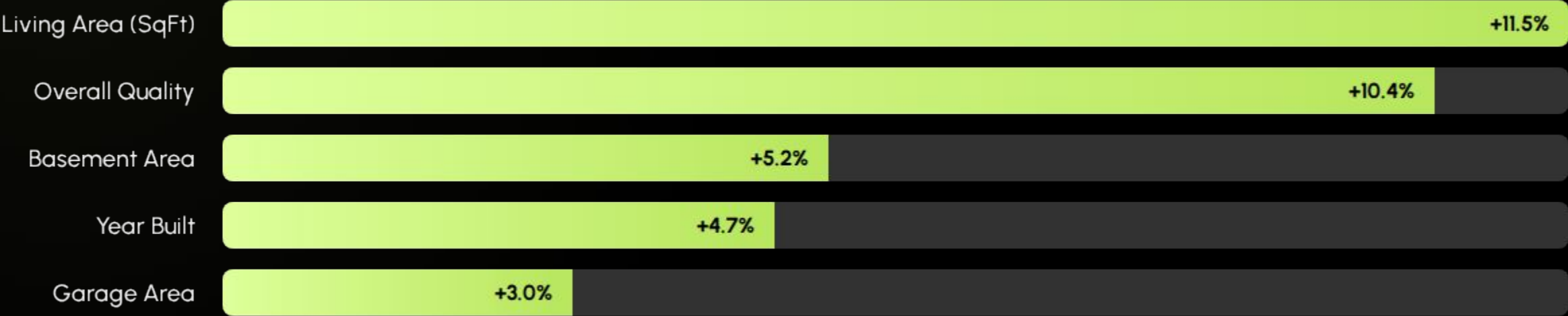
Visualizing Success

In our final tests, predicted prices tracked actual sale prices with incredible consistency, forming a tight "Cigar Shape" that confirms model reliability across all price brackets.

The Top Drivers of Value

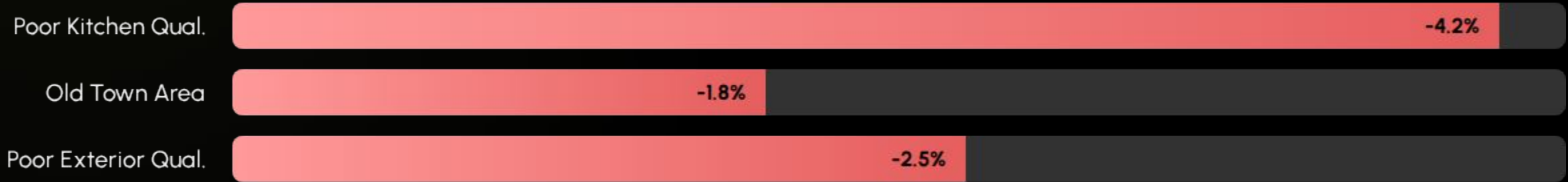
Where should you invest your renovation dollars?

Top Value Adders



*Percentages represent the estimated impact on total home value per unit increase.

Value Shrinkers to Avoid



Identifying "Red Flag" features that actively pull down a property's market potential.

Model Comparison

Valuation Method	Accuracy (R ²)	Complexity	Business Verdict
Linear Regression	0.8925	High (Noise)	Prone to minor errors
Ridge Regression	0.8926	Moderate	Stable, but busy
Lasso Regression	0.8932	Optimized	Selected: Best Precision

Investor Takeaways

- ✔ **Focus on Space:** Square footage is still king, adding ~11% value for every major standard deviation increase.
- ✔ **Quality Matters:** Improving the "Overall Quality" score is the second most profitable move.
- ✔ **Location Premium:** Neighborhoods like Stone Brook and Northridge command significant data-backed premiums.



THANK YOU

Any Questions?

Leveraging Data Science for Smarter Real Estate Decisions.

Ames Housing Market Analytics | <https://shaulamarquez.github.io/portfolio/>

Technical Methodology & Reproducibility



Github

Link: <https://github.com/shaulamarquez/Aimes-Housing-Marketing-Analysis>



Aimes Housing Data Set

Link: https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-ML0232EN-SkillsNetwork/asset/Ames_Housing_Data1.tsv
